

Manishansh Shaswat

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EDUCATION

Singapore University of Technology and Design, Singapore

Sep 22 to Present

- Bachelor of Engineering (Computer Science and Design)
- Date of graduation: May 2026

WORK EXPERIENCES (LATEST)

Diagnostics Development Hub (DxD) Hub, A*STAR

Singapore
Oct 24 to Jan 25

- App Development: Built a dynamic mental health application in Flutter using GetX, Flutter Modular, and BLoC. Implemented seamless animations via AnimationController with custom Bezier curves. Ensured cross-platform consistency with Material & Cupertino design.
- User Experience: Designed a responsive pre-login survey with intuitive navigation (PageView/ListView) and dynamic progress indicators for enhancing user engagement.
- Backend & Prototyping: Contributed to telemetry for gait analysis by utilising Grafana Loki and OpenTelemetry. Integrated APIs for data retrieval with the frontend segment of the application with dio (a Flutter Dart package, equivalent to axios in JavaScript). Leveraged Flutter Flow for reusable widget classes to execute rapid iterations.
- Cloud Architecture: Assisted in designing an AWS-based business admin UI flow diagram consisting of AWS components (Lambda, EKS, CodePipeline) for scalable system integration.

TRU Technologies LLC

Dubai, UAE
Dec 23 to Jan 24

- Worked under the guidance of the TRU team on the basics of Enterprise Resource Planning (ERP) software business solutions, which improved workflow efficiency by 16%
- Performed the role of data analytics by examining data relevant to ERP

ACADEMIC PROJECTS (LATEST)

Singapore University of Technology and Design, Singapore

50.039 Theory and Practice of Deep Learning US Interest Rate Training Model

Feb 25 to Apr 25

- Built a custom Seq2Seq model using PyTorch to forecast US interest rate from historical time series data, achieving MAE of 0.39 and F1 score of 0.94
- Processed economic datasets with 80-10-10 train-validation-test split; final model reached 88% accuracy with $\pm 1\%$ margin
- Documented the training pipeline and used pickle to save and load model weights, ensuring full reproducibility

50.038 Computational Data Science Bull to Bear Tesla Stock Transition Prediction

Feb 25 to Apr 25

- Developed an LSTM classifier for predicting bull-to-bear Tesla stock market transition using tweet sentiment analysis achieving a F1 score of 0.88
- Benchmarked model against Random Forest (F1 Score: 0.96) and Gradient Boosting (F1 Score: 0.97) for sentiment-driven forecasting
- Automated a full pipeline to clean tweets, merge Tesla stock data, and generate predictions on over 1,200 daily values

50.021 Artificial Intelligence

Feb 25 to Apr 25

- Engineered a hate speech classification NLP system using Pytorch and fine-tuned distilRoBERTA, obtaining 87.4% test accuracy on a balanced dataset of over 700,000 social media samples
- Benchmarked around 5+ transformer models (e.g. RoBERTA, distilBERT) for analysis of tradeoffs in performance and overfitting
- Deployed a Streamlit GUI (web-page application) for hate speech filtering, enabling live classification of input text as "Hateful" or "Non-Hateful"

CERTIFICATIONS

Machine Learning Specialization from Stanford

Dec 24 to Feb 25

- Supervised Machine Learning: Regression and Classification
- Advanced Learning Algorithms
- Unsupervised Learning, Recommender, Reinforcement Learning

Generative AI for Software Development from DeepLearning.AI

Nov 24 to Dec 24

- Introduction to Generative AI for Software Development
- Team Software Engineering with AI
- AI-Powered Software and System Design

HACKATHONS & COMPETITIONS

- What the Hack (WTH), Singapore, 2022
- YouthXHack Total Defence Hackathon, Singapore, 2023
- I Participated in the RoboRoarZ National Competition held at the Singapore University of Technology & Design

ADDITIONAL INFORMATION

- Technical Skills: Java, Python, HTML, XML, Alchitry Labs, CSS, Bootstrap, Flutter, OpenTelemetry, Express.js, Logistic Regression, Linear Regression, Reinforcement Learning AWS Cloud Architecture, React.js, Node.js, Machine Learning, Lucid Programming, Grafana Loki